



**ĐẠI HỌC
BÁCH KHOA HÀ NỘI**
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Research and Detection of Attacks on the Siemens S7Comm Protocol in ICS/OT Networks

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ONE LOVE. ONE FUTURE.



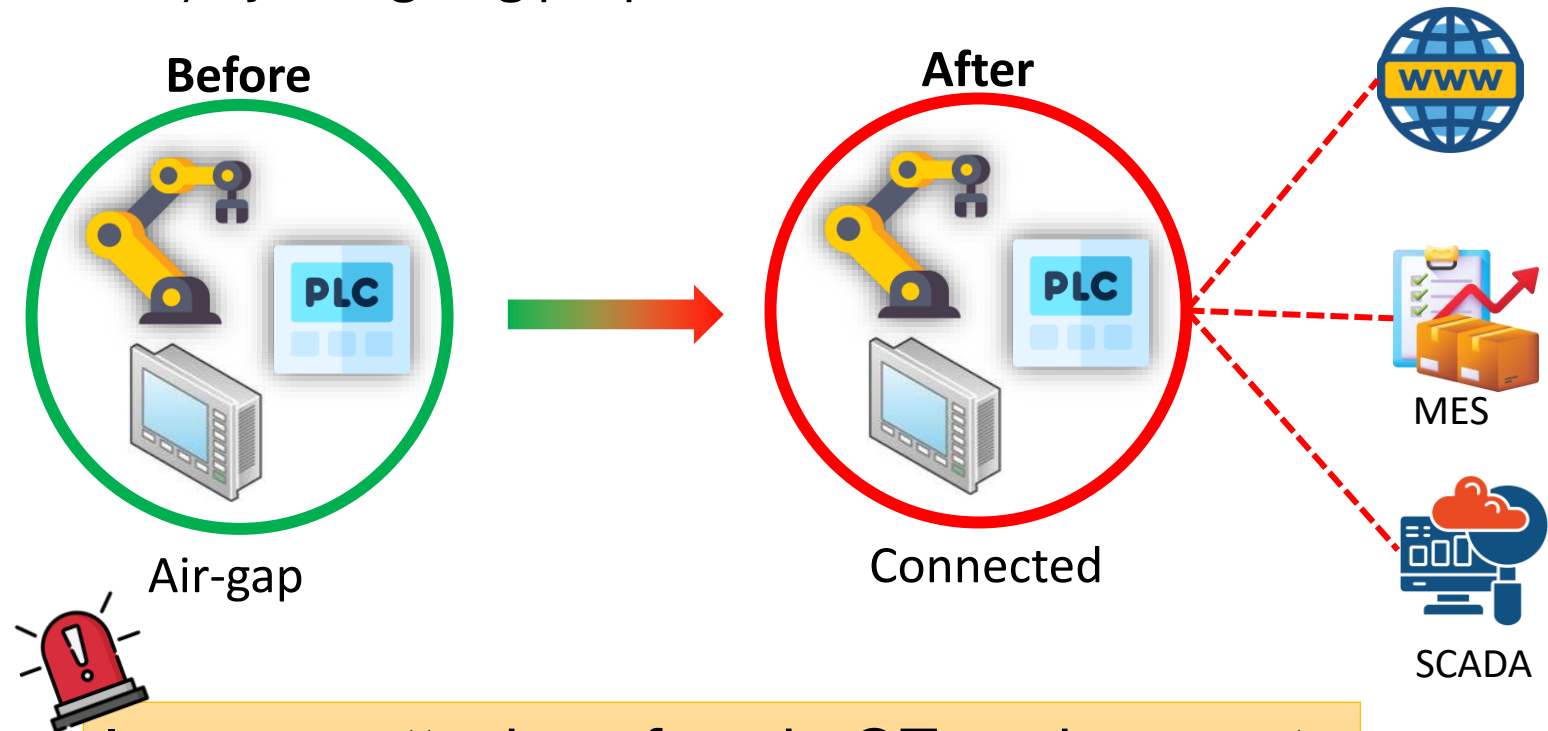
Outline

- Objectives
- Background theory
- Problem analysis
- System & methodology
- Experimental results
- Conclusion



Security risks in OT

ICS protocols are often proprietary (for reliability), not include **encryption**, **authentication**, or **integrity** mechanisms (for real-time performance) by design purposes



Increase attack surface in OT environments

**ICS (Industrial Control Systems) is any hardware and software systems used to monitor and control physical industrial processes and infrastructure.*

Objectives

Challenges:

- Increasing number of attacks with a tendency toward nation-state and political motivations
- Factories operate continuously, making it impossible to redesign from scratch using a "secure by design" philosophy. **Availability** is the highest priority.
- Intervening in industrial devices to install active monitoring agents is extremely risky (device lifespan, proprietary software, vendor diversity, etc.)

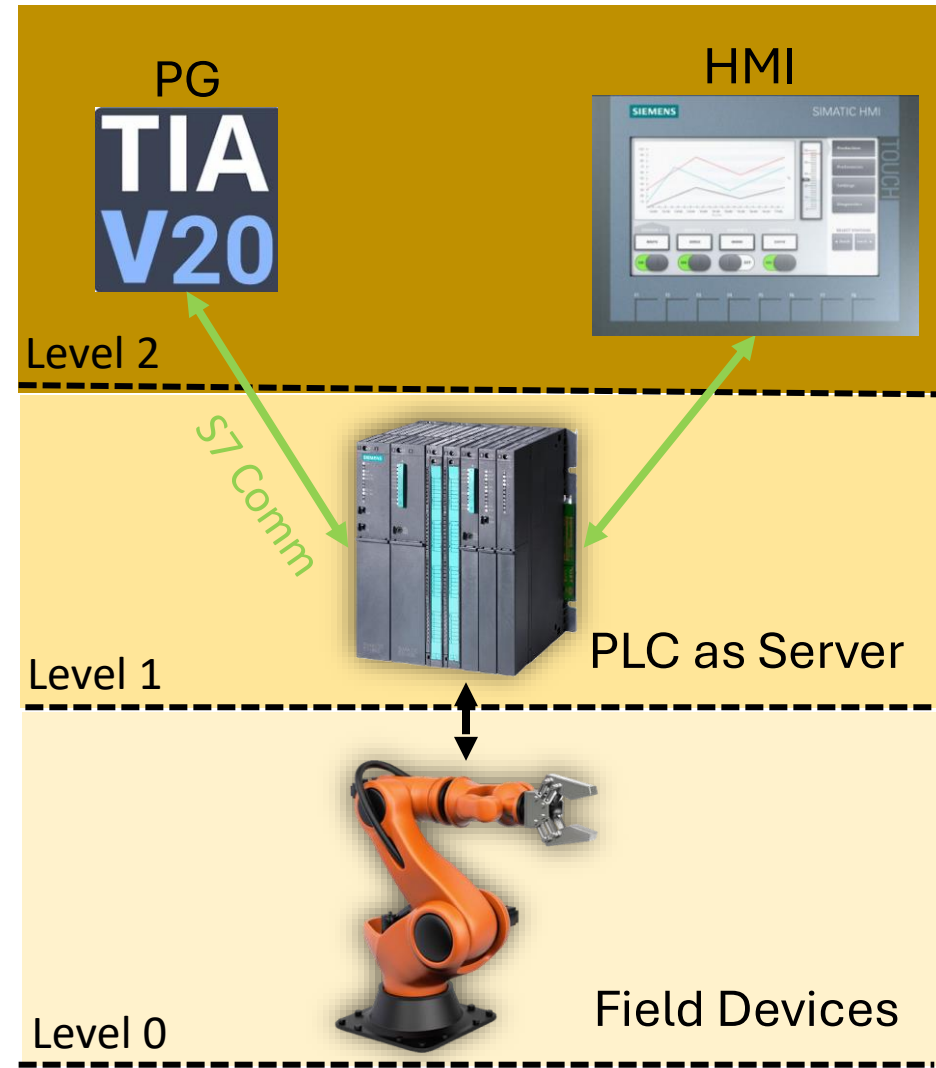


Building a passive monitoring solution specified for industrial protocol to minimize the impact on factory operations

Background theory

- Proprietary Siemens protocol for programming, monitoring, and control Siemens ecosystem over TCP port 102, using request – response model
- S7 Comm: used on PLC S7-200/300/400 devices with:
 - **no authentication**
 - **no authorization**
 - **no encryption**

**PLC (Programmable Logic Controller) is a rugged industrial computer that continuously monitors inputs, executes control logic, and controls outputs in real time.*



S7 Comm in Purdue model IEC 62443

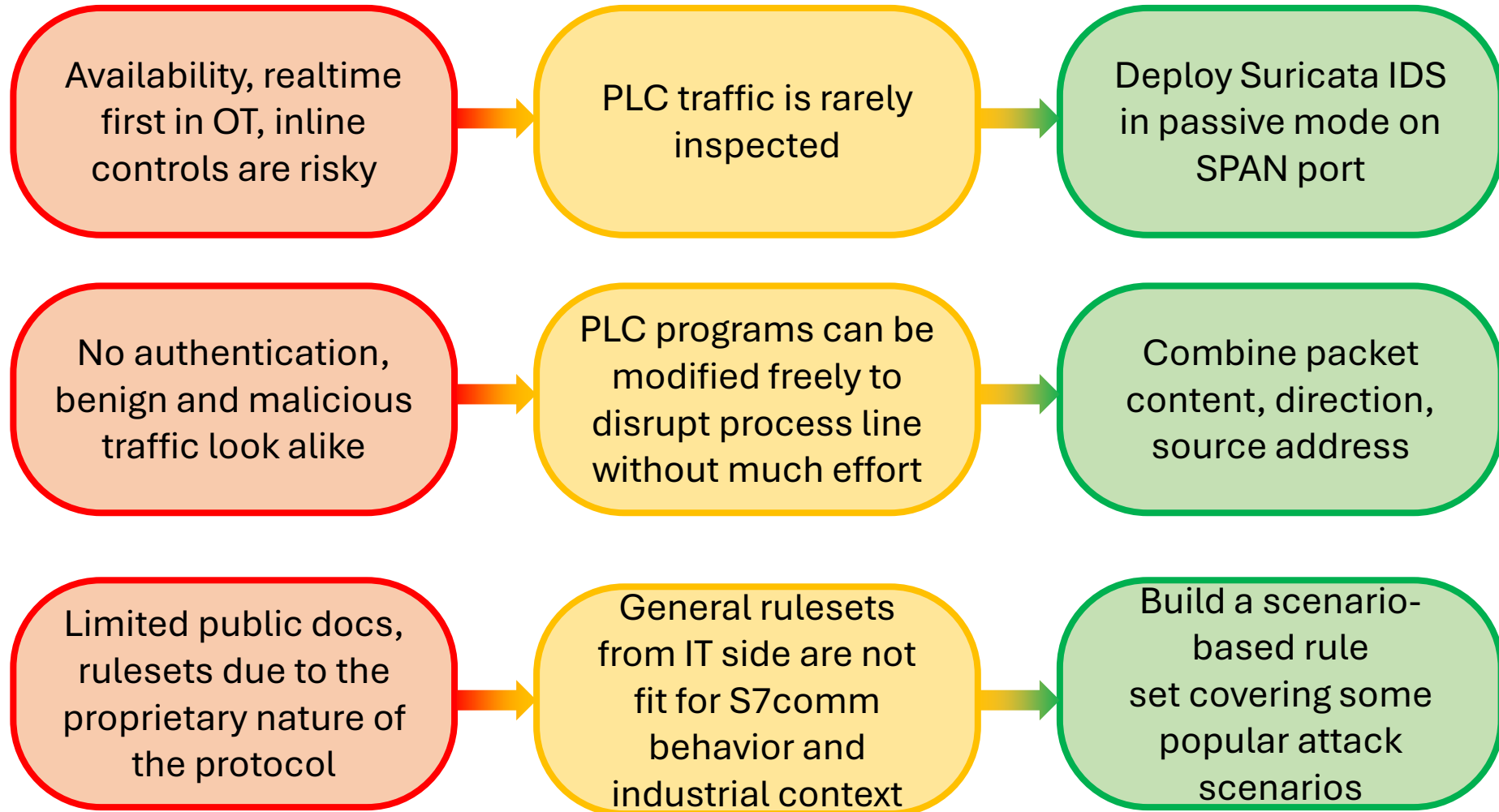
Background theory

- **IDS** (Intrusion Detection System): monitor network traffic to alert on abnormal behaviors or unauthorized access.
- Categorize by detection method:
 - **Signatures-based**: Using predefined rule → Stable, low false positive
 - Anomaly-based: Compare against normal baseline → Detect 0-day, high false positive



SURICATA®

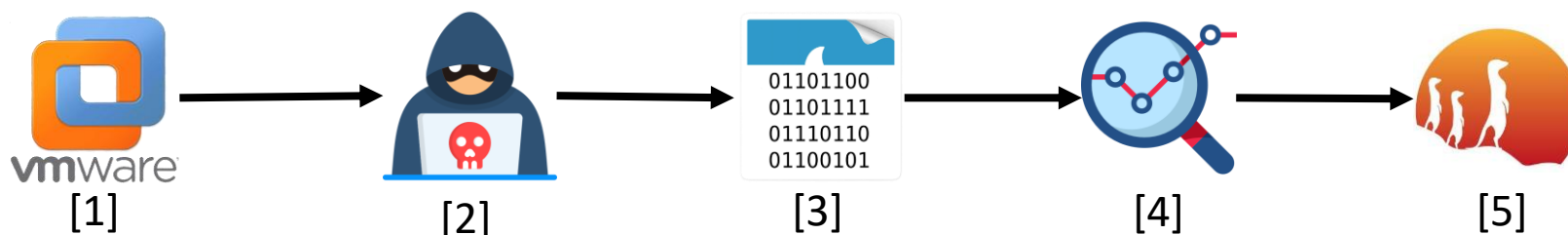
Problem analysis



System & methodology

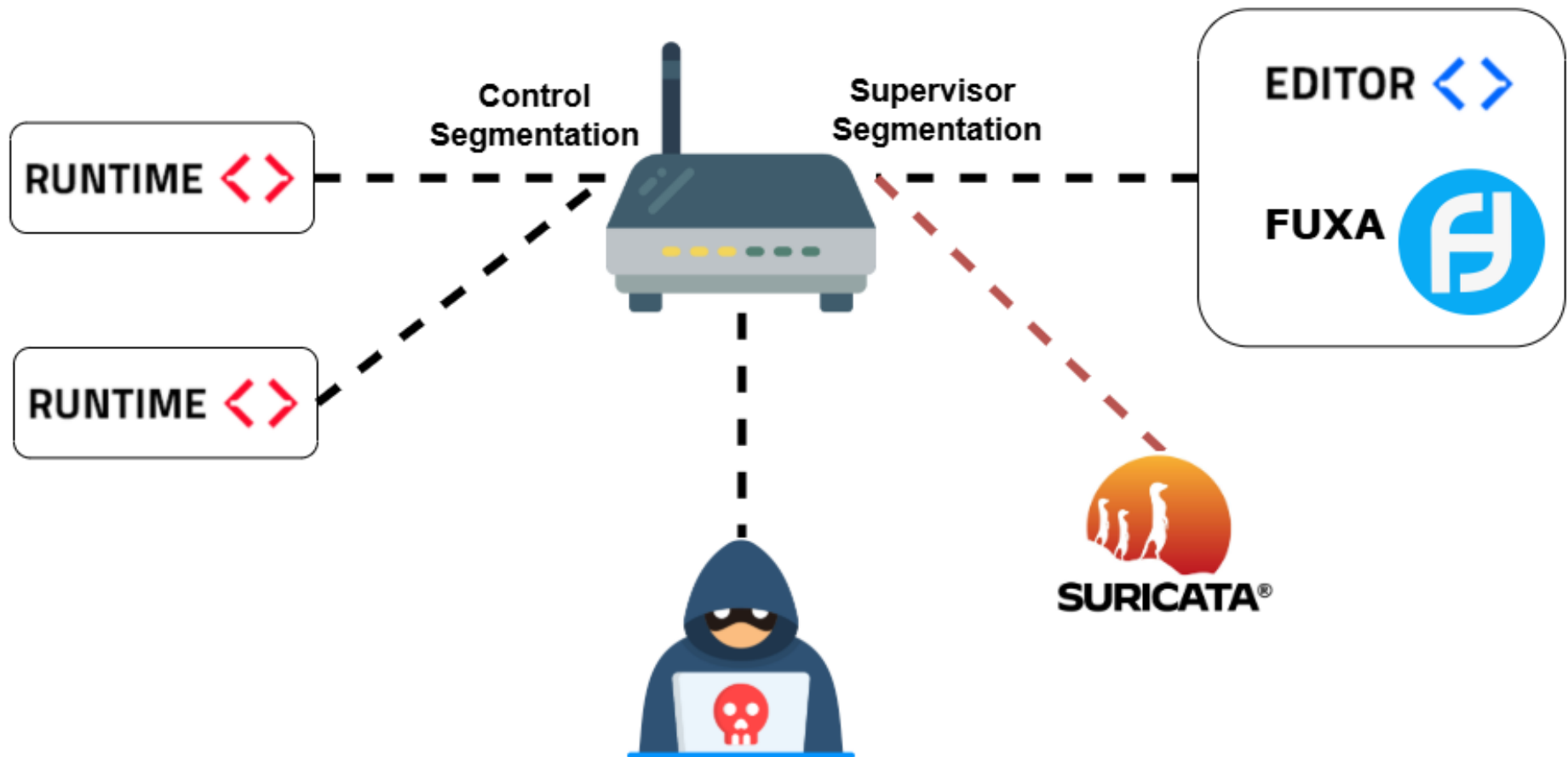
Overview approach

1. **Deploy simulation:** deploy the simulation environment on VMware
2. **Building attack modules:** using Snap 7 and tools to build attack scenarios
3. **Traffic collection and analysis:** capture network traffic with Wireshark, analyze S7comm packet structure
4. **Detection rule development:** convert observed indicators into Suricata signatures based on byte offsets, function codes, frequency thresholds, and abnormal syntax.
5. **IDS testing:** run Suricata on traffic copies and compare alerts with each simulated scenario.



System & methodology

Deploy simulation



Simulation network topology

System & methodology

Building attack modules

Category	Details	Risk
Reconnaissance	Discover PLC services through TCP/102 scanning, SZL reading, and block enumeration	Expose PLC assets and support later exploitation
Process manipulation	Modify process variables using unauthorized WriteVar and spoofed ReadVar operations	Alter process values and mislead operators
Program transfer	Unauthorized upload and download of PLC program blocks	Steal or replace PLC control logic
Command injection	Abuse PLC control functions by sending Start/Stop commands	Halt or restart production without authorization
Denial of Service	Overload PLC communication using high-rate S7 functions and TCP SYN flooding	Disrupt monitoring and control communication
Malformed S7comm packets	Crafts structurally invalid TPKT/COTP/S7	Destabilize PLC communication and cause service interruption

System & methodology

Traffic collection and analysis

192.168.1.10	192.168.1.40	TCP	66 4305 → 102 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM		TCP Handshake
192.168.1.40	192.168.1.10	TCP	60 102 → 4305 [SYN, ACK] Seq=0 Ack=1 Win=4096 Len=0 MSS=1460		
192.168.1.10	192.168.1.40	TCP	54 4305 → 102 [ACK] Seq=1 Ack=1 Win=64240 Len=0		COTP
192.168.1.10	192.168.1.40	COTP	76 CR TPDU src-ref: 0x000f dst-ref: 0x0000		
192.168.1.40	192.168.1.10	COTP	76 CC TPDU src-ref: 0x0003 dst-ref: 0x000f		Init S7
192.168.1.10	192.168.1.40	S7COMM	79 ROSCTR:[Job] Function:[Setup communication]		
192.168.1.40	192.168.1.10	S7COMM	81 ROSCTR:[Ack Data] Function:[Setup communication]		S7 actual data exchange
192.168.1.10	192.168.1.40	COTP	61 DT TPDU (0) [COTP fragment, 0 bytes]		
192.168.1.10	192.168.1.40	S7COMM	87 ROSCTR:[Userdata] Function:[Request] -> [CPU functions] -> [Read SZL] ID=0x0132 Index=0x0004		
192.168.1.40	192.168.1.10	S7COMM	135 ROSCTR:[Userdata] Function:[Response] -> [CPU functions] -> [Read SZL] ID=0x0132 Index=0x0004		
192.168.1.10	192.168.1.40	COTP	61 DT TPDU (0) [COTP fragment, 0 bytes]		
192.168.1.10	192.168.1.40	S7COMM	87 ROSCTR:[Userdata] Function:[Request] -> [CPU functions] -> [Read SZL] ID=0x0000 Index=0x0000		
192.168.1.40	192.168.1.10	S7COMM	301 ROSCTR:[Userdata] Function:[Response] -> [CPU functions] -> [Read SZL] (S7COMM fragment id=5)		
192.168.1.10	192.168.1.40	COTP	61 DT TPDU (0) [COTP fragment, 0 bytes]		
192.168.1.10	192.168.1.40	S7COMM	87 ROSCTR:[Userdata] Function:[Request] -> [CPU functions] -> [Read SZL]		
192.168.1.40	192.168.1.10	S7COMM	105 ROSCTR:[Userdata] Function:[Response] -> [CPU functions] -> [Read SZL] (S7COMM reassembled id=5) ID=0x0000 Index=0x0000		

- **Purpose:** Collected traffic traces are inspected to identify protocol-level patterns and attack indicators.
- **Analysis focus:**
 1. Separate normal communication layers: TCP, COTP, and S7comm
 2. Observe request/response structure and packet sequences
 3. Extract stable fields such as function code, ROSCTR, PDU type, and payload length
 4. Compare benign and attack traces to define detection features

Detection rule development

- **General approach**
 - Signature-based detection for S7comm attack behaviors
 - Use protocol anchors from TPKT, COTP, and S7comm headers
 - Match operation-specific indicators: ROSCTR, function code, PDU type
 - Use thresholds for repeated traffic such as DoS
- **Rule grouping** by SID range:
 - **100000x**: session setup and reconnaissance
 - **100001x**: read/write variables
 - **100002x**: block upload/download
 - **100003x**: Start/Stop control commands
 - **100004x**: DoS and rate-based floods
 - **100005x**: malformed S7comm PDUs

System & methodology

Detection rule development

```
18 0.006558205 192.168.60.20 192.168.50.10 S7COMM 99 ROSCTR:[Userdata] Function:[Request] -> [CPU functions] -> [Read S...
> Frame 18: 99 bytes on wire (792 bits), 99 bytes captured (792 bits) on interface ens33, id 0
> Ethernet II, Src: VMware_20:46:7c (00:50:56:20:46:7c), Dst: VMware_f1:b8:5e (00:0c:29:f1:b8:5e)
> Internet Protocol Version 4, Src: 192.168.60.20, Dst: 192.168.50.10
> Transmission Control Protocol, Src Port: 52900, Dst Port: 102, Seq: 48, Ack: 50, Len: 33
> TPKT, Version: 3, Length: 33
  Version: 3
  Reserved: 0
  Length: 33
> ISO 8073/X.224 COTP Connection-Oriented Transport Protocol
  Length: 2
  PDU Type: DT Data (0x0f)
  [Destination reference: 0x30000]
  .000 0000 = TPDU number: 0x00
  1... .... = Last data unit: Yes
> S7 Communication
  > Header: (Userdata)
  > Parameter: (Request) ->(CPU functions) ->(Read SZL)
  > Data (SZL-ID: 0x0011, Index: 0x0001)
```

```
0000 00 0c 29 f1 b8 5e 00 50 56 20 46 7c 08 00 45 00 ..).^P V F |..E.
0010 00 55 4e ad 40 00 40 06 fc 86 c0 a8 3c 14 c0 a8 .UN.@.@. ....<..
0020 32 0a ce a4 00 66 d9 92 b2 99 9d c8 3c dc 80 18 2...f... ..<...
0030 01 f6 ff 14 00 00 01 01 08 0a 69 5b 97 bc ca 96 .....i[....
0040 7e 28 03 00 00 21 02 f0 80 32 07 00 00 00 00 00 ~(...!.. .2....
0050 08 00 08 00 01 12 04 11 44 01 00 ff 09 00 04 00 .....D...
0060 11 00 01
```

Example rule

```
alert tcp !$SUPERVISOR_NET any -> $CONTROL_NET 102 (msg:"Read SZL ...";
```

```
flow:to_server,established; content:"|03 00|"; offset:0; depth:2; content:"|02 F0 80 32
```

```
07|"; offset:4; depth:5; content:"|00 01 12 04 11 44 01 00|"; offset:17; depth:8;
```

```
sid:1000002; rev:1; priority:2;)
```

Tools

- **Open PLC**: open-source platform that simulates a physical PLC, supports 5 standard control programs under the IEC 61131-3
 - Open PLC Runtime V4
 - Open PLC Editor v4
- **FUXA**: open-source, web-based HMI that supports Siemens S7comm protocol
- **Snap7**: an open-source library developed to connect to, transmit data with, and control automation devices through popular industrial protocol, including S7 Comm.

RUNTIME 
EDITOR 



Aspect	Metric
Detection coverage	Ratio = detected scenarios / total scenarios
Sensor capacity	Decoded packets and kernel_drops under load

- **Detection criterion:** A scenario is counted as *detected* when at least one expected rule (SID) is triggered while its traffic is present.
- **Scope:** 10 representative attack classes: reconnaissance, process read/write , block transfer, DoS, malformed PDUs, PLC Start/Stop
- **Interpretation:** IDS processes traffic without dropping packets before decoding, especially during DoS.

Experimental results

Scenario	Expected SIDs	Outcome
Reconnaissance	1000001–1000005	Met
Process Read/Write	1000010, 1000011, 1000013	Met
Block transfer	1000020–1000026	Met
Command injection	1000030–1000033	Met
Denial of service	1000040–1000043	Met
Malformed PDUs	1000050–1000058	Met

- **Detection coverage:** All 10 attack classes met the detection criterion, at least one expected rule fired for each scenario.
- **Sensor capacity:** Kernel drops = 0 throughout the campaign, including the DoS phase . The passive sensor processed mirrored traffic without packet loss.

Conclusion

- **Main contributions**
 - Built an ICS lab for S7comm traffic generation and analysis
 - Implemented representative attack modules for reconnaissance, manipulation, DoS, and malformed traffic
 - Designed a structured Suricata rule set using protocol signatures and scenario context
 - Evaluated detection coverage, baseline alerts, and IDS processing load
- **Conclusion:** Passive IDS monitoring is feasible for the studied S7comm, but real deployment requires site-specific tuning and operational context
- **Future Improvement**
 - Validate on real Siemens PLC hardware
 - Correlate with SIEM, SOAR, NAC
 - Test larger traffic loads and more complex plant topologies
 - Extend detection to S7comm-plus environments

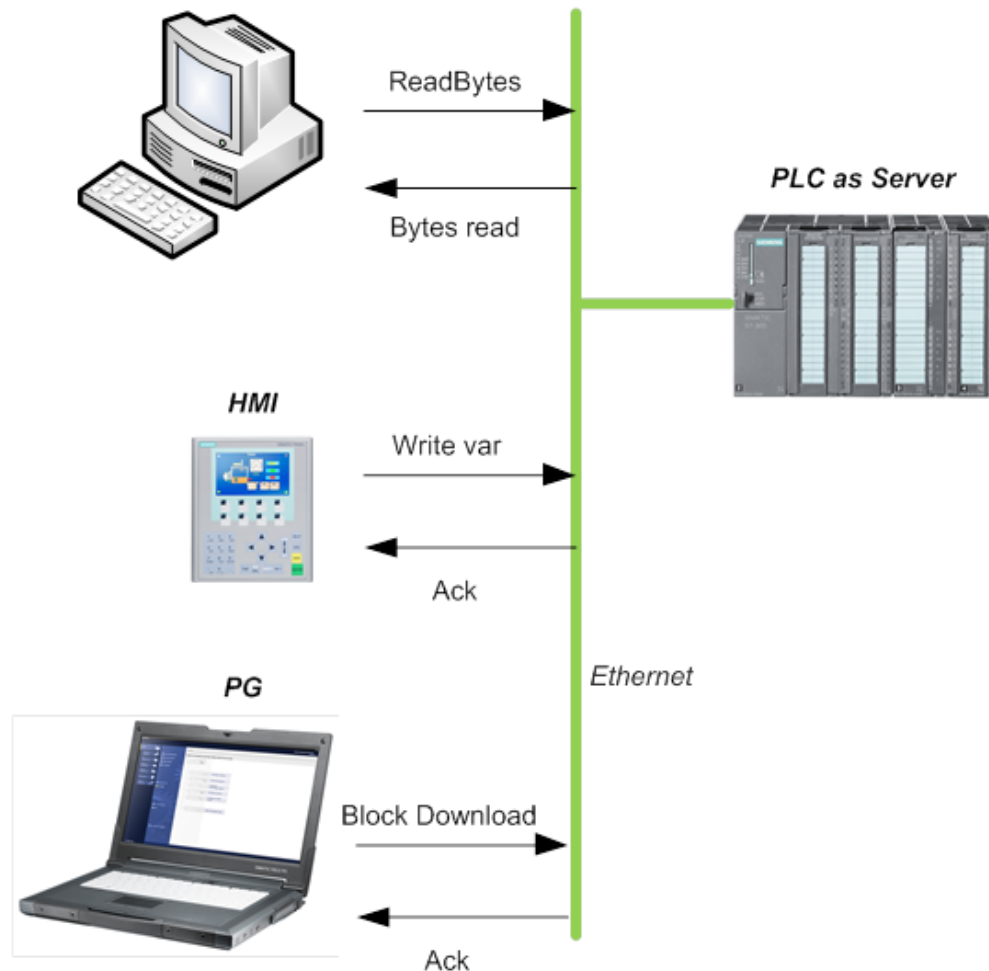
A decorative graphic on the left side of the slide. It features a dark blue background with a large, stylized circular pattern composed of many small red dots. The dots are arranged in concentric, slightly irregular rings, creating a sense of depth and movement. The word "HUST" is written in white, bold, sans-serif capital letters, centered within the blue area.

HUST

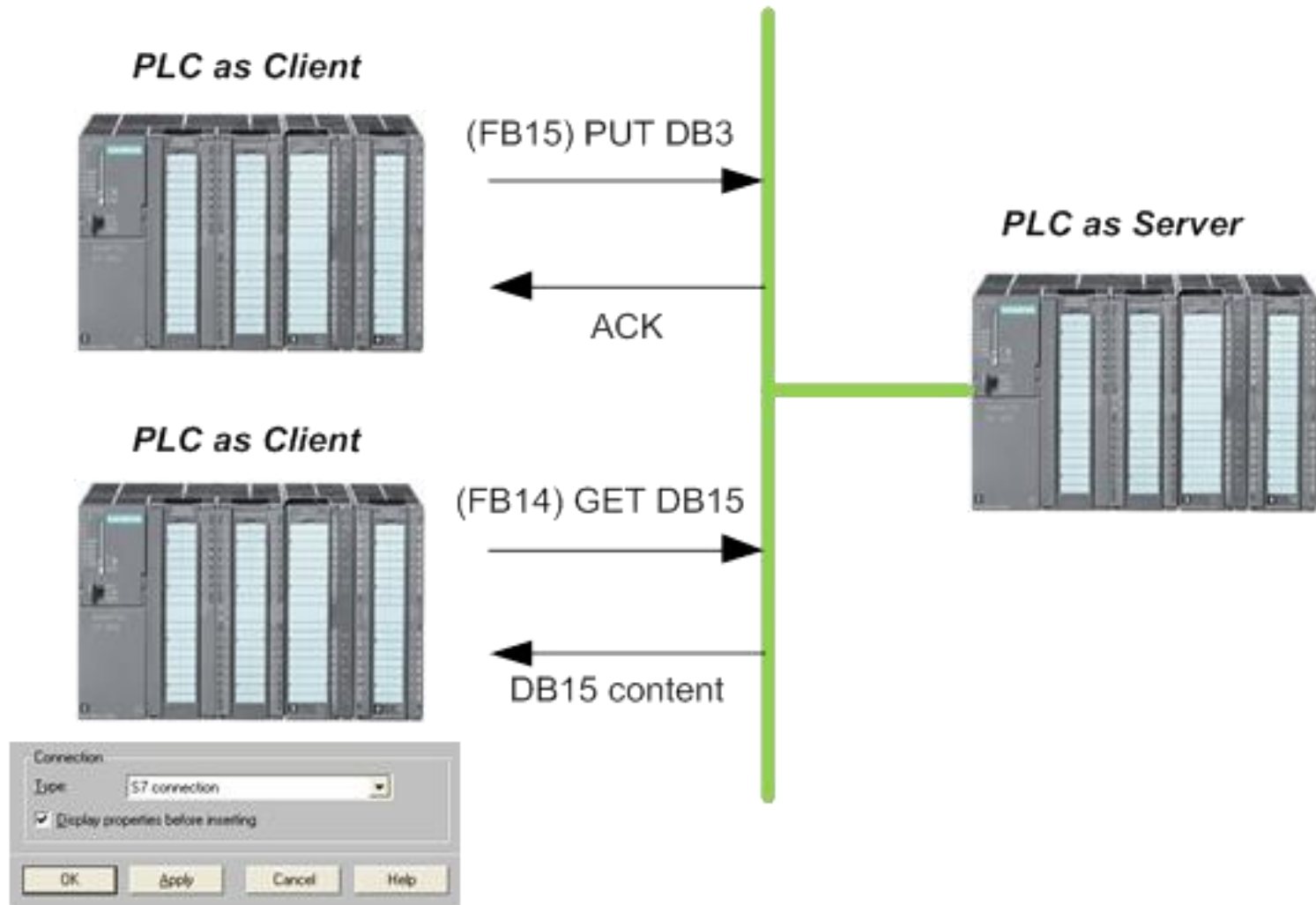
**THANK YOU FOR
YOUR
ATTENTION !**

S7 Components

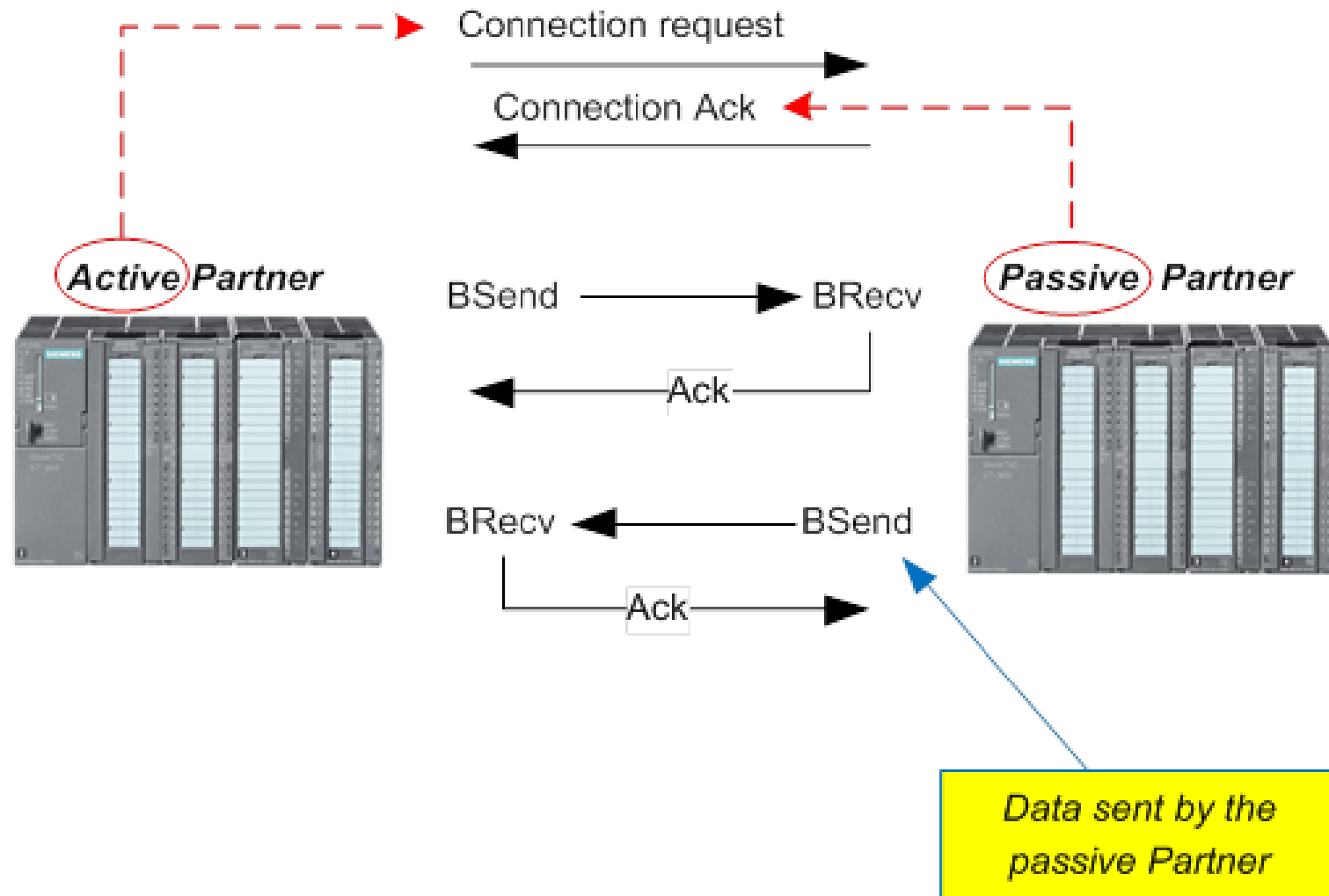
- *LibNoDave*
- *Simatic NET*
- *OPC Server*
- *Snap7 Client*



S7 Components

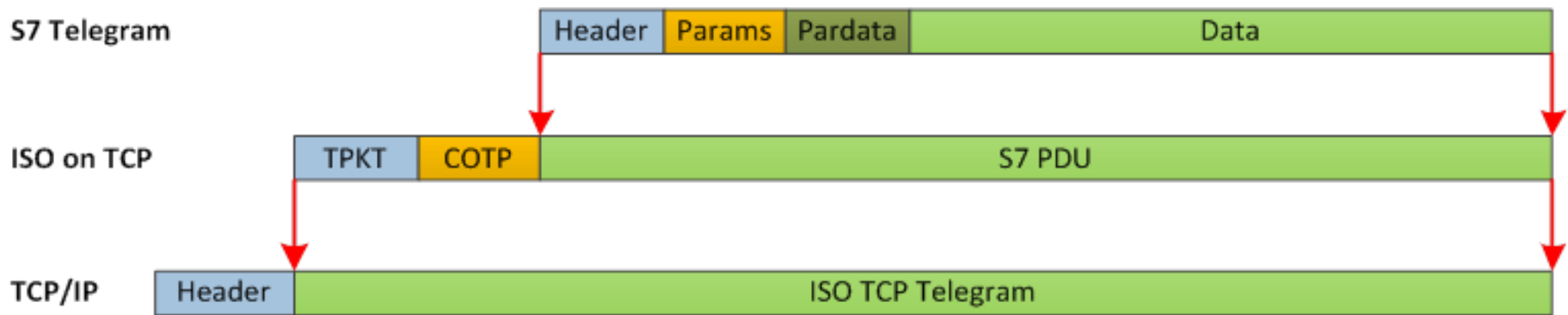


S7 Components

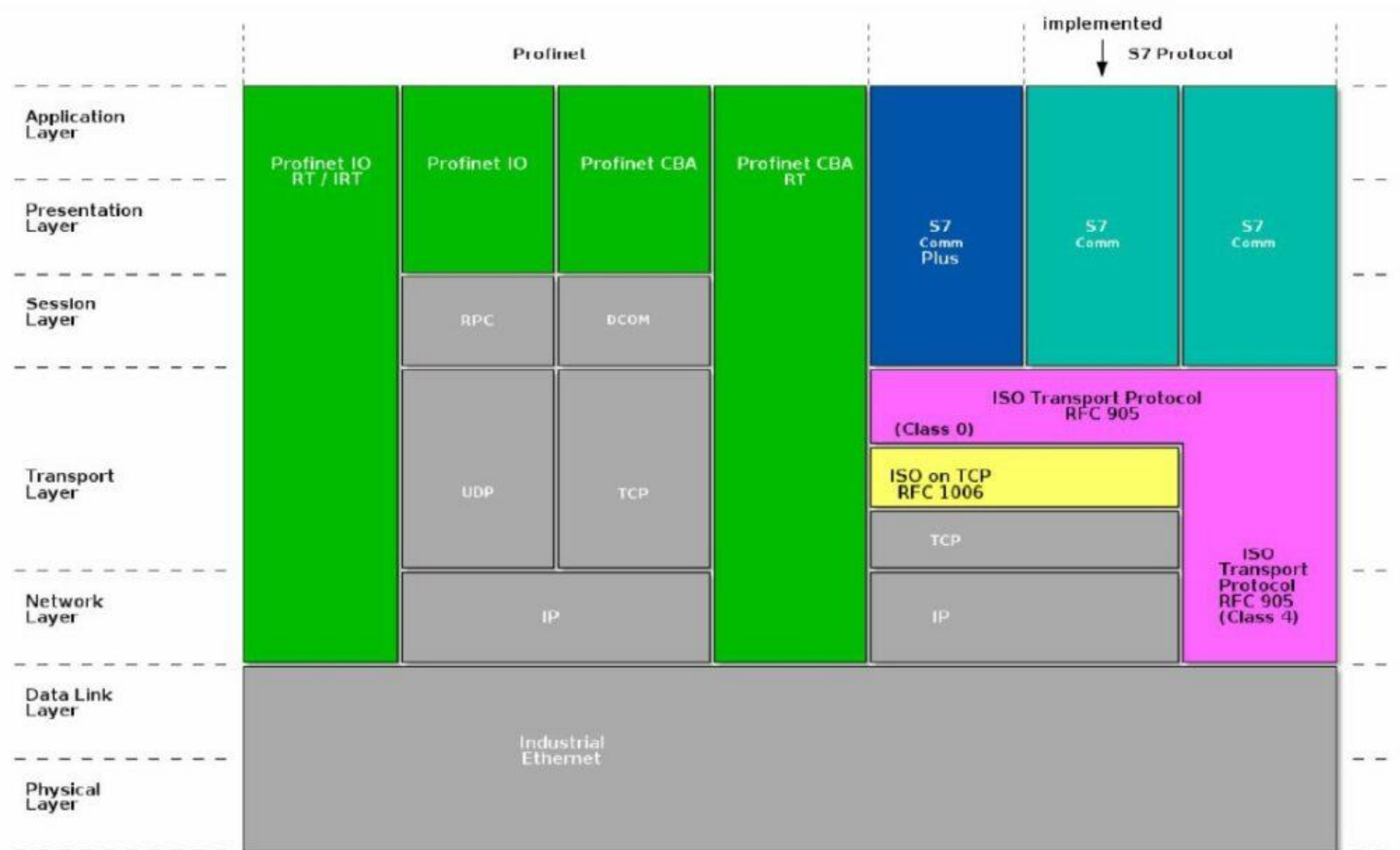


S7 Protocol Stack

- Runs on **ISO-on-TCP** over Profinet cable. Block-oriented protocol. Each block is called a PDU
- COTP establishes logical connection (compatible with TCP connection-oriented)
- TPKT wrapper fixed-length package (compatible with TCP stream-based)



S7 Protocol Stack



Threshold

×

Device Property

Name

PLC_1

Type

SiemensS7

Polling

1 sec

Enable

☒

IP-Address (IP 127.0.0.1)

192.168.50.10

Rack

0

Slot

2

CANCEL

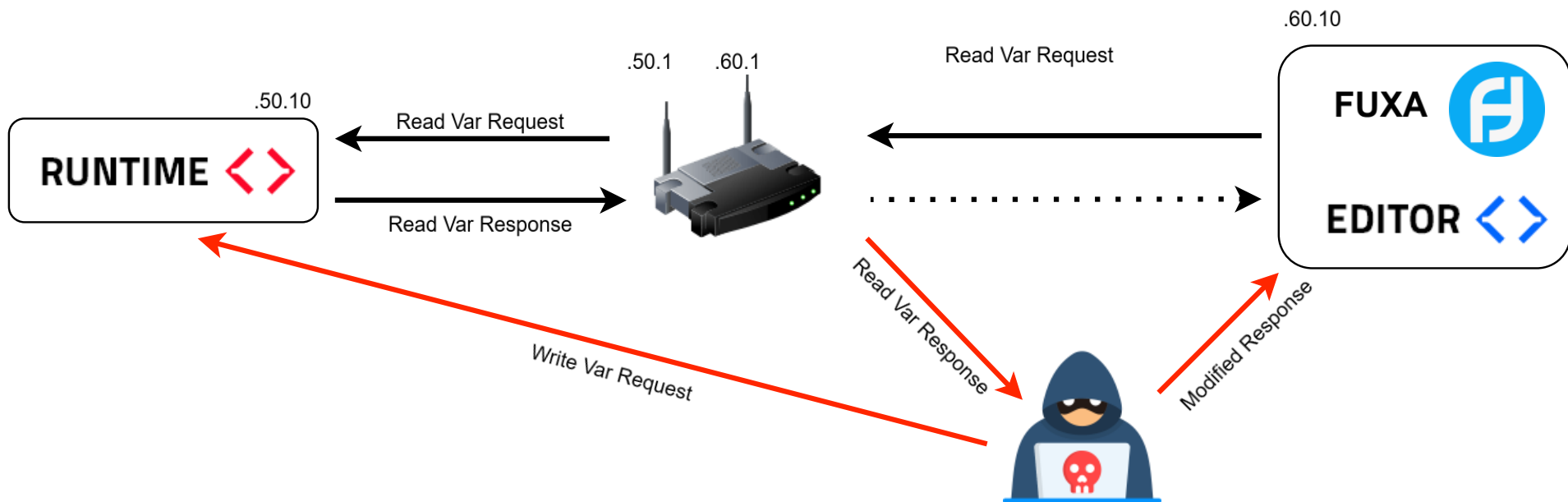
OK

S7 Protocol Stack

Các yếu tố có thể ảnh hưởng đến việc chọn Threshold:

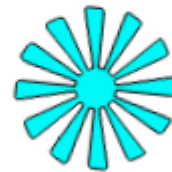
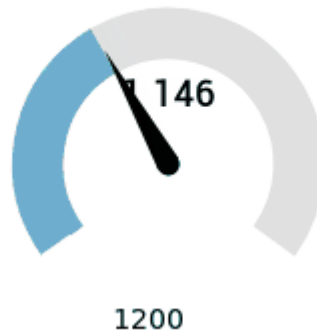
- Số lượng client, hiệu năng xử lý của các thiết bị, của hệ thống mạng hiện tại
- Chu kỳ polling và số tags: HMI poll mỗi bao nhiêu s, mỗi cycle đọc bao nhiêu tag
- Cách các thiết bị phần cứng, firmware khác nhau xử lý. Ví dụ PLC S7-300/400 chịu tải ít hơn S7-1200/1500
- Context làm việc của riêng xưởng đó. Đặc biệt thay đổi khi có đợt bảo dưỡng hoặc thi công mới

Stuxnet



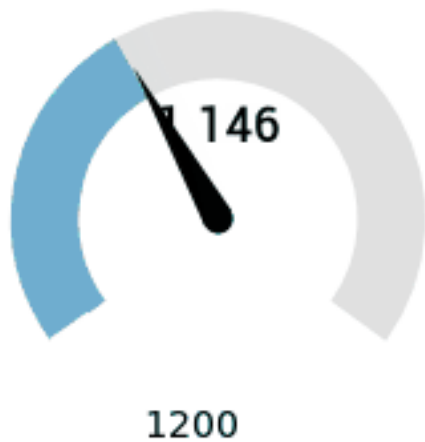
Stuxnet

Device		Filter							
← PLC_1								⋮ ↺	
+	Name	Address	Device	Type	Value	Timestamp	Description	✕	
✎	iSimSpeed	db1.dbw0	PLC_1	Int	1101	15-04-2026 12:15:04		⋮	✕
✎	iSetpoint	db1.dbw2	PLC_1	Int	1200	15-04-2026 12:15:04		⋮	✕
✎	bCentrifugeStatudb2.dbb0		PLC_1	Bool	0	15-04-2026 12:15:04		⋮	✕



Stuxnet

Device		Filter							
PLC_1									
+	Name	Address	Device	Type	Value	Timestamp	Description		
	iSimSpeed	db1.dbw0	PLC_1	Int	1101	15-04-2026 12:15:04			
	iSetpoint	db1.dbw2	PLC_1	Int	1200	15-04-2026 12:15:04			
	bCentrifugeStatudb2.dbb0		PLC_1	Bool	0	15-04-2026 12:15:04			



Console

Debugger

PLC Logs

Variables

main.iSimSpeed

INT 1083

main.iSetpoint

INT 1200

main.iCounter

INT 542

main.bCentrifugeStatus

BOOL FALSE

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.60.10	192.168.50.10	S7COMM	97	ROSCTR:[Job] Function:[Read Var]
2	0.001472623	192.168.50.10	192.168.60.10	S7COMM	95	ROSCTR:[Ack_Data] Function:[Read Var]

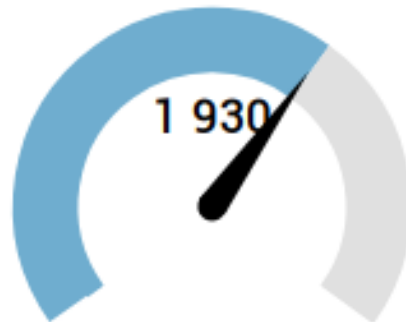
- ▶ Frame 2: 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface ens33, id 0
- ▶ Ethernet II, Src: VMware_5d:57:17 (00:0c:29:5d:57:17), Dst: VMware_2f:0d:bb (00:0c:29:2f:0d:bb)
- ▶ Internet Protocol Version 4, Src: 192.168.50.10, Dst: 192.168.60.10
- ▶ Transmission Control Protocol, Src Port: 102, Dst Port: 38508, Seq: 1, Ack: 32, Len: 29
- ▶ TPkt, Version: 3, Length: 29
- ▶ ISO 8073/X.224 COTP Connection-Oriented Transport Protocol
- ▼ S7 Communication
 - ▶ Header: (Ack_Data)
 - ▼ Parameter: (Read Var)
 - Function: Read Var (0x04)
 - Item count: 1
 - ▼ Data
 - ▼ Item [1]: (Success)
 - Return code: Success (0xff)
 - Transport size: BYTE/WORD/DWORD (0x04)
 - Length: 4
 - Data: 042d04b0

Biến iSimSpeed là 042d do nó nằm ở trước trong cấu hình các PLC tags, biến iSetpoint là 04b0. S7 dùng Little Endian, nên giá trị 042d tương ứng với 1069 và 04b0 tương ứng với 1200

Stuxnet

Current Speed		Setpoint Speed		Centrifuge Status
b'\x03\xc1'		b'\x04\xb0'		b'\x00'
b'\x07\xe4'		b'\x07\xd0'		b'\x01'

🛡️ 📄 localhost:1881/lab



2000



Stuxnet

```
▼ S7 Communication
  ▶ Header: (Ack_Data)
  ▼ Parameter: (Read Var)
    Function: Read Var (0x04)
    Item count: 1
  ▼ Data
    ▼ Item [1]: (Success)
      Return code: Success (0xff)
      Transport size: BYTE/WORD/DWORD (0x04)
      Length: 4
      Data: 043304b0
```

Giá trị `iSetpoint` trên gói tin tới HMI đang là 1200 (04b0)

Console Debugger PLC Logs

Variables

- main.iSimSpeed INT 1982
- main.iSetpoint INT 2000
- main.iCounter INT 438
- main.bCentrifugeStatus BOOL TRUE

Range 10 seconds

Kiểm tra debugger của OpenPLC Editor cũng cho thấy thực tế máy đang chạy giao động quanh mức 2000

Recon

```
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-15 14:55 UTC
Nmap scan report for 192.168.50.10
Host is up (0.0021s latency).
PORT      STATE SERVICE
102/tcp   open  iso-tsap
| s7-info:
|   Module: 6ES7 315-2EH14-0AB0
|   Basic Hardware: 6ES7 315-2EH14-0AB0
|   Version: 3.2.6
|   System Name: SNAP7-SERVER
|   Module Type: CPU 315-2 PN/DP
|   Serial Number: S C-C2UR28922012
|_  Copyright: Original Siemens Equipment
Service Info: Device: specialized
Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
=== Finished (recon_nmap) exit=0 ===
```

```
07/15/2026-10:55:46.950752  [**] [1:1000001:2] ICS S7COMM Setup
Communication to PLC [**] [Classification: (null)] [Priority: 3] {TCP}
192.168.60.20:51208 -> 192.168.50.10:102
07/15/2026-10:55:46.952605  [**] [1:1000002:2] ICS S7COMM Read SZL
request - PLC information enumeration [**] [Classification: (null)]
[Priority: 2] {TCP} 192.168.60.20:51208 -> 192.168.50.10:102
07/15/2026-10:55:46.952605  [**] [1:1000003:2] ICS S7COMM Read SZL module
identification 0x0011 [**] [Classification: (null)] [Priority: 2] {TCP}
192.168.60.20:51208 -> 192.168.50.10:102
```




```
cd "/home/ubuntu/Thesis" && "/home/ubuntu/Thesis/.venv/bin/python"  
"attacks/reconnaissance/scanBlock.py" --ip 192.168.50.10 --rack 0  
--slot 2 --list-only 2>&1  
All blocks: <block list count OB: 0 FB: 0 FC: 0 SFB: 0 SFC: 0x0 DB:  
2 SDB: 0>  
DB 1: (list-only, no get_block_info)  
DB 2: (list-only, no get_block_info)  
=== Finished (recon_blocks) exit=0 ===
```

```
07/15/2026-11:02:52.747294  [**] [1:1000005:3] ICS S7COMM List Blocks  
request - PLC block inventory enumeration [**] [Classification: (null)]  
[Priority: 2] {TCP} 192.168.60.20:33390 -> 192.168.50.10:102  
07/15/2026-11:02:52.748925  [**] [1:1000005:3] ICS S7COMM List Blocks  
request - PLC block inventory enumeration [**] [Classification: (null)]  
[Priority: 2] {TCP} 192.168.60.20:33390 -> 192.168.50.10:102
```

Write Var

```
cd "/home/ubuntu/Thesis" && "/home/ubuntu/Thesis/.venv/bin/python"  
"attacks/stuxnet_mitm_sim/attacker/test_readwrite_db_via_snap7.py"  
--ip 192.168.50.10 --rack 0 --slot 2 2>&1  
Current Speed | Setpoint Speed | Centrifuge Status  
-----  
1045          |      1200          |      0  
1896          |      2000          |      1  
=== Finished (write_db) exit=0 ===
```

```
07/15/2026-11:05:08.942031  [**] [1:1000010:2] ICS S7COMM Write Var  
request to PLC [**] [Classification: (null)] [Priority: 1] {TCP}  
192.168.60.20:38890 -> 192.168.50.10:102  
07/15/2026-11:05:08.942031  [**] [1:1000011:3] ICS S7COMM Write Var to DB  
memory area [**] [Classification: (null)] [Priority: 1] {TCP}  
192.168.60.20:38890 -> 192.168.50.10:102
```

All DoS ruleset using 10s-time windows, count by source (track by_src), only count for traffic that is outside **supervisor** (!\$SUPERVISOR_NET):

SID	DoS Type	Threshold
1000040	TCP SYN flood to port 102	40 SYN / 10s
1000041	Setup Communication flood	25 Setup PDU / 10s
1000042	Read SZL flood	25 Read SZL / 10s
1000043	Generic S7 Job flood	80 Job PDU / 10s

```
07/15/2026-11:06:29.220733  [**] [1:1000001:2] ICS S7COMM Setup  
Communication to PLC [**] [Classification: (null)] [Priority: 3] {TCP}  
192.168.60.20:43462 -> 192.168.50.10:102  
07/15/2026-11:06:29.220286  [**] [1:1000001:2] ICS S7COMM Setup  
Communication to PLC [**] [Classification: (null)] [Priority: 3] {TCP}  
192.168.60.20:43474 -> 192.168.50.10:102  
07/15/2026-11:06:29.220514  [**] [1:1000001:2] ICS S7COMM Setup  
Communication to PLC [**] [Classification: (null)] [Priority: 3] {TCP}  
192.168.60.20:43468 -> 192.168.50.10:102  
07/15/2026-11:06:29.222293  [**] [1:1000041:1] ICS S7COMM possible Setup  
Communication flood to PLC [**] [Classification: (null)] [Priority: 1]  
{TCP} 192.168.60.20:51238 -> 192.168.50.10:102
```

```
07/15/2026-11:06:58.323086  [**] [1:1000002:2] ICS S7COMM Read SZL  
request - PLC information enumeration [**] [Classification: (null)]  
[Priority: 2] {TCP} 192.168.60.20:38981 -> 192.168.50.10:102  
07/15/2026-11:06:58.324026  [**] [1:1000002:2] ICS S7COMM Read SZL  
request - PLC information enumeration [**] [Classification: (null)]  
[Priority: 2] {TCP} 192.168.60.20:38981 -> 192.168.50.10:102  
07/15/2026-11:06:58.324846  [**] [1:1000042:1] ICS S7COMM possible Read  
SZL flood - PLC service load [**] [Classification: (null)] [Priority: 1]  
{TCP} 192.168.60.20:38983 -> 192.168.50.10:102
```

```
07/15/2026-11:14:25.616589  [**] [1:1000050:2] ICS S7COMM invalid TPKT  
version on ISO-TSAP [**] [Classification: (null)] [Priority: 2] {TCP}  
192.168.60.20:42408 -> 192.168.50.10:102
```

```
07/15/2026-11:14:34.432642  [**] [1:1000051:2] ICS S7COMM TPKT length  
exceeds TCP payload [**] [Classification: (null)] [Priority: 2] {TCP}  
192.168.60.20:60838 -> 192.168.50.10:102
```

```
07/15/2026-11:14:43.247504  [**] [1:1000058:3] ICS S7COMM TPKT length  
smaller than TCP payload [**] [Classification: (null)] [Priority: 2]  
{TCP} 192.168.60.20:38612 -> 192.168.50.10:102
```

```
07/15/2026-11:14:50.060469  [**] [1:1000056:4] ICS S7COMM invalid COTP DT  
header toward PLC [**] [Classification: (null)] [Priority: 2] {TCP}  
192.168.60.20:53948 -> 192.168.50.10:102
```

```
07/15/2026-11:14:58.876792  [**] [1:1000052:2] ICS S7COMM invalid S7  
protocol id [**] [Classification: (null)] [Priority: 2] {TCP}  
192.168.60.20:45602 -> 192.168.50.10:102
```

```
07/15/2026-11:15:08.695766  [**] [1:1000053:1] ICS S7COMM invalid ROSCTR  
value [**] [Classification: (null)] [Priority: 2] {TCP}  
192.168.60.20:34890 -> 192.168.50.10:102
```

```
07/15/2026-11:15:16.507706  [**] [1:1000054:1] ICS S7COMM invalid Job  
function code [**] [Classification: (null)] [Priority: 2] {TCP}  
192.168.60.20:34918 -> 192.168.50.10:102
```

```
07/15/2026-11:15:25.322649  [**] [1:1000055:2] ICS S7COMM truncated S7  
Job PDU toward PLC [**] [Classification: (null)] [Priority: 2] {TCP}  
192.168.60.20:33656 -> 192.168.50.10:102
```


Start/Stop

```
cd "/home/ubuntu/Thesis" && "/home/ubuntu/Thesis/.venv/bin/python"  
"attacks/start_stop_plc/start_stop_plc.py" 192.168.50.10 stop  
--rack 0 --slot 2 2>&1  
[+] S7 Ack OK for stop (function 0x29)  
=== Finished (plc_stop) exit=0 ===
```

```
07/15/2026-11:18:40.957005  [**] [1:1000032:2] ICS S7COMM PLC Stop  
command [**] [Classification: (null)] [Priority: 1] {TCP}  
192.168.60.20:49604 -> 192.168.50.10:102  
07/15/2026-11:18:40.957005  [**] [1:1000033:2] ICS S7COMM replayed Stop  
CPU P_PROGRAM [**] [Classification: (null)] [Priority: 1] {TCP}  
192.168.60.20:49604 -> 192.168.50.10:102
```

Start/Stop

```
cd "/home/ubuntu/Thesis" && "/home/ubuntu/Thesis/.venv/bin/python"  
"attacks/start_stop_plc/start_stop_plc.py" 192.168.50.10 start  
--rack 0 --slot 2 2>&1  
[+] S7 Ack OK for start (function 0x28)  
=== Finished (plc_start) exit=0 ===
```

```
07/15/2026-11:19:48.813000  [**] [1:1000030:2] ICS S7COMM PLC Control  
command [**] [Classification: (null)] [Priority: 1] {TCP}  
192.168.60.20:57748 -> 192.168.50.10:102  
07/15/2026-11:19:48.813000  [**] [1:1000031:2] ICS S7COMM replayed Start  
CPU P_PROGRAM [**] [Classification: (null)] [Priority: 1] {TCP}  
192.168.60.20:57748 -> 192.168.50.10:102
```


Download Upload

```
lubuntu@lubuntu2204:~$ sudo suricata -r /home/lubuntu/Desktop/s7comm_downloading_block_db1.pcap -c /etc/suricata/suricata.yaml -k none -l /var/log/suricata --set stream.midstream=true
15/7/2026 -- 11:24:06 - <Notice> - This is Suricata version 6.0.4 RELEASE running in USER mode
15/7/2026 -- 11:24:06 - <Notice> - all 3 packet processing threads, 4 management threads initialized, engine started.
15/7/2026 -- 11:24:06 - <Notice> - Signal Received. Stopping engine.
15/7/2026 -- 11:24:06 - <Notice> - Pcap-file module read 1 files, 93 packets, 8011 bytes
```

```
08/20/2014-06:00:03.969906  [**] [1:1000020:3] ICS S7COMM Request
Download block to PLC [**] [Classification: (null)] [Priority: 1] {TCP}
192.168.1.10:4185 -> 192.168.1.40:102
08/20/2014-06:00:03.977452  [**] [1:1000021:3] ICS S7COMM PLC requests
Download Block data [**] [Classification: (null)] [Priority: 1] {TCP}
192.168.1.40:102 -> 192.168.1.10:4185
08/20/2014-06:00:03.977707  [**] [1:1000026:3] ICS S7COMM client sends
Download Block data to PLC [**] [Classification: (null)] [Priority: 1]
{TCP} 192.168.1.10:4185 -> 192.168.1.40:102
08/20/2014-06:00:03.982421  [**] [1:1000021:3] ICS S7COMM PLC requests
Download Block data [**] [Classification: (null)] [Priority: 1] {TCP}
192.168.1.40:102 -> 192.168.1.10:4185
08/20/2014-06:00:03.982566  [**] [1:1000026:3] ICS S7COMM client sends
Download Block data to PLC [**] [Classification: (null)] [Priority: 1]
{TCP} 192.168.1.10:4185 -> 192.168.1.40:102
08/20/2014-06:00:03.987375  [**] [1:1000021:3] ICS S7COMM PLC requests
Download Block data [**] [Classification: (null)] [Priority: 1] {TCP}
192.168.1.40:102 -> 192.168.1.10:4185
08/20/2014-06:00:03.987573  [**] [1:1000026:3] ICS S7COMM client sends
Download Block data to PLC [**] [Classification: (null)] [Priority: 1]
{TCP} 192.168.1.10:4185 -> 192.168.1.40:102
```